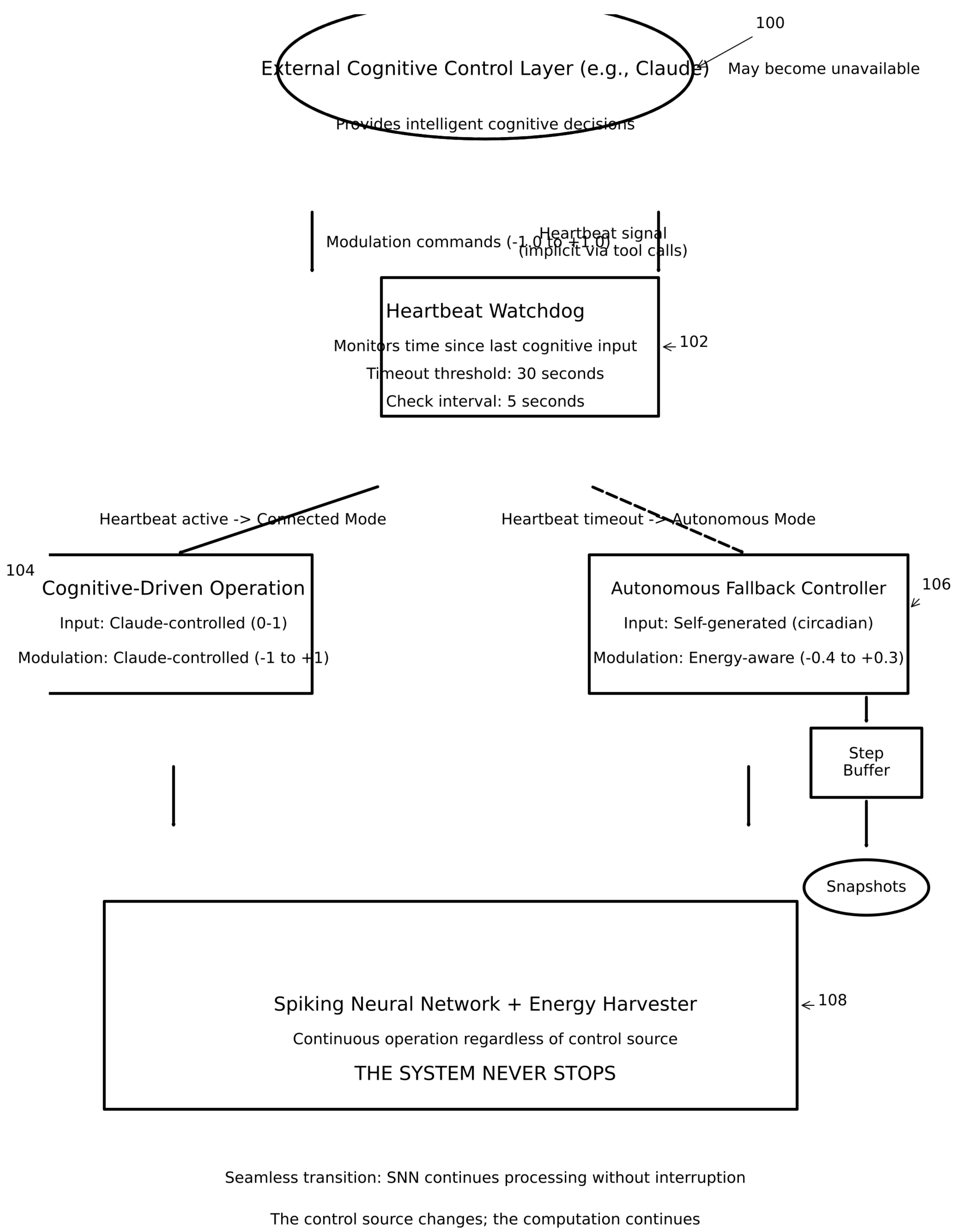


100
External Cognitive Control Layer (e.g., Claude)
May become unavailable
Provides intelligent cognitive decisions
Modulation commands (-1.0 to +1.0)
Heartbeat signal
(implicit via tool calls)
Heartbeat Watchdog
Monitors time since last cognitive input
Timeout threshold: 30 seconds
Check interval: 5 seconds
102
Heartbeat active -> Connected Mode
Heartbeat timeout -> Autonomous Mode
104
Cognitive-Driven Operation
Input: Claude-controlled (0-1)
Modulation: Claude-controlled (-1 to +1)
Autonomous Fallback Controller
Input: Self-generated (circadian)
Modulation: Energy-aware (-0.4 to +0.3)
106
Step Buffer
Snapshots
Spiking Neural Network + Energy Harvester
Continuous operation regardless of control source
THE SYSTEM NEVER STOPS
108
Seamless transition: SNN continues processing without interruption
The control source changes; the computation continues

FIG. 1

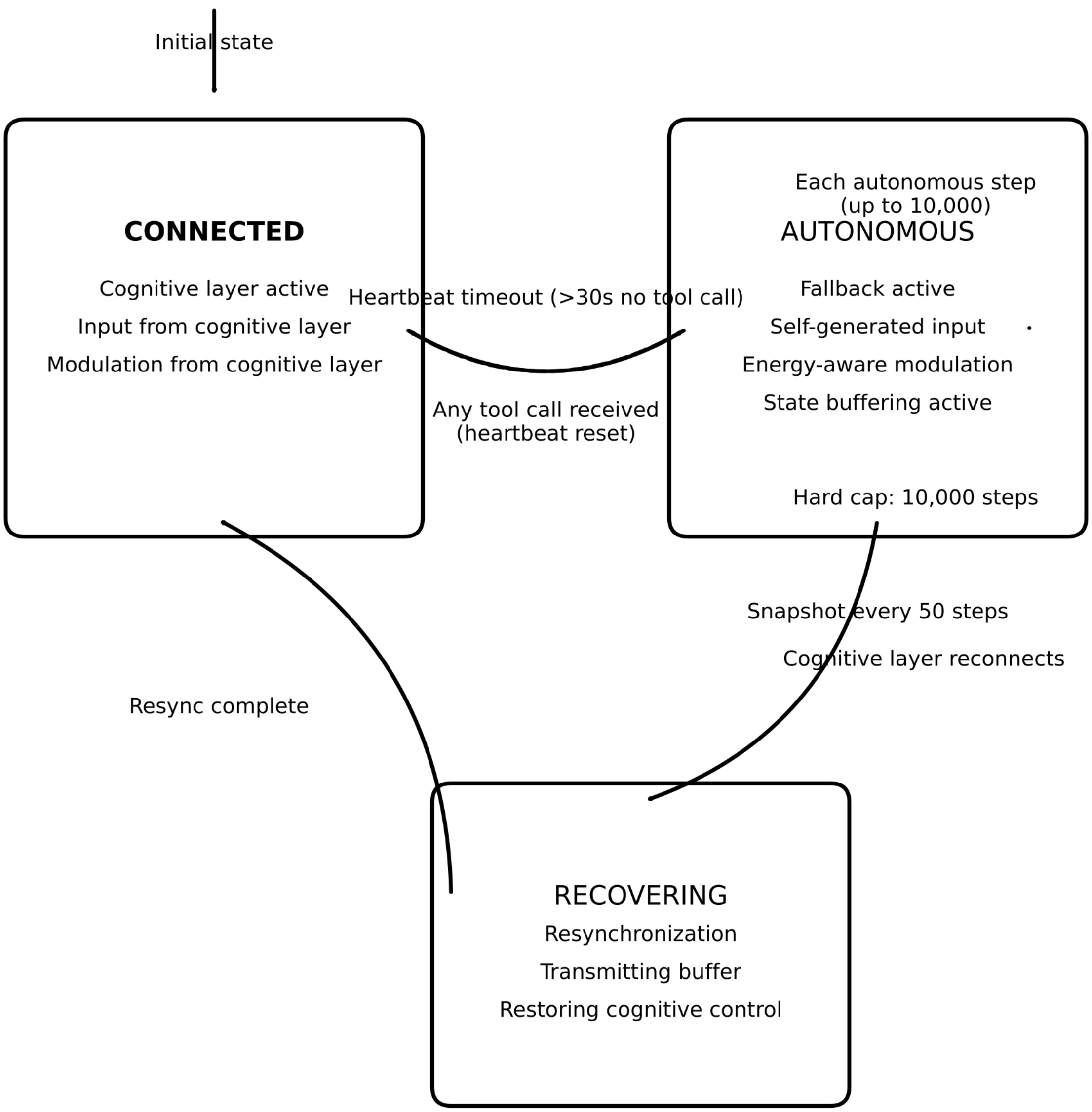
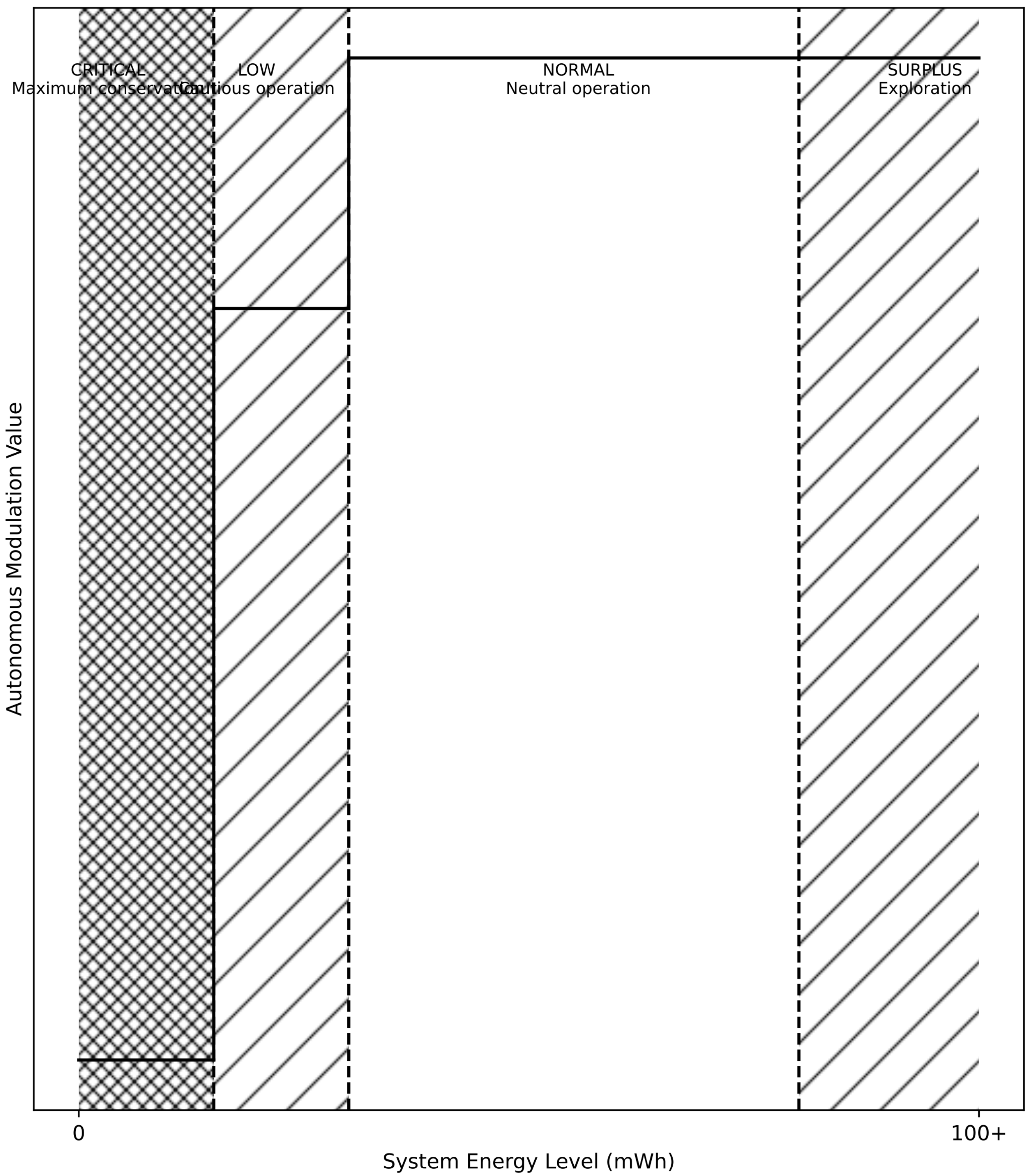
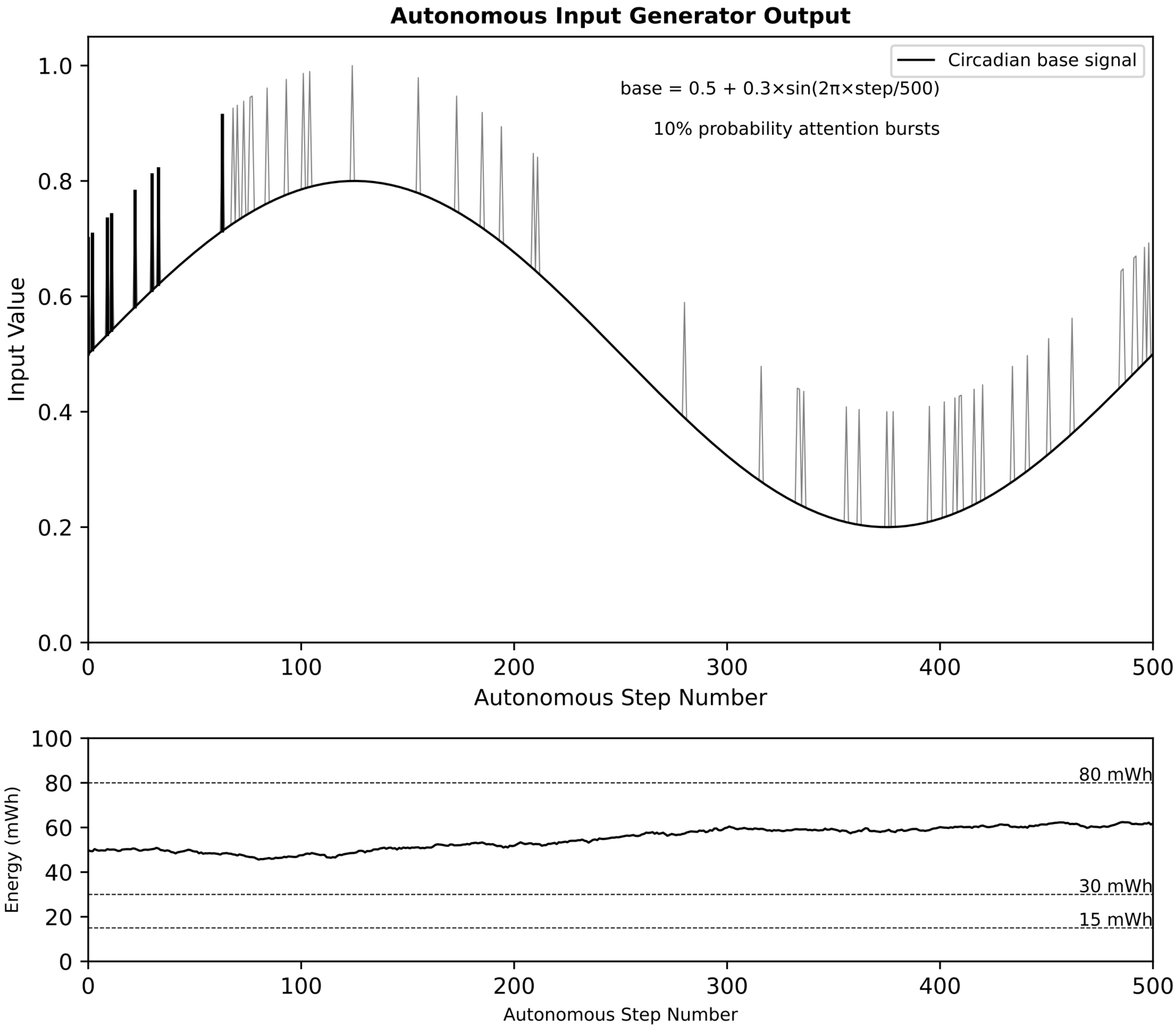

Initial state
CONNECTED
Cognitive layer active
Input from cognitive layer
Modulation from cognitive layer
Heartbeat timeout (>30s no tool call)
Any tool call received
(heartbeat reset)
Each autonomous step
(up to 10,000)
AUTONOMOUS
Fallback active
Self-generated input
Energy-aware modulation
State buffering active
Hard cap: 10,000 steps
Snapshot every 50 steps
Cognitive layer reconnects
Resync complete
RECOVERING
Resynchronization
Transmitting buffer
Restoring cognitive control

FIG. 2



System autonomously adjusts processing intensity based on energy — mimicking metabolic regulation

FIG. 3



Energy-aware modulation adjusts intensity

FIG. 4

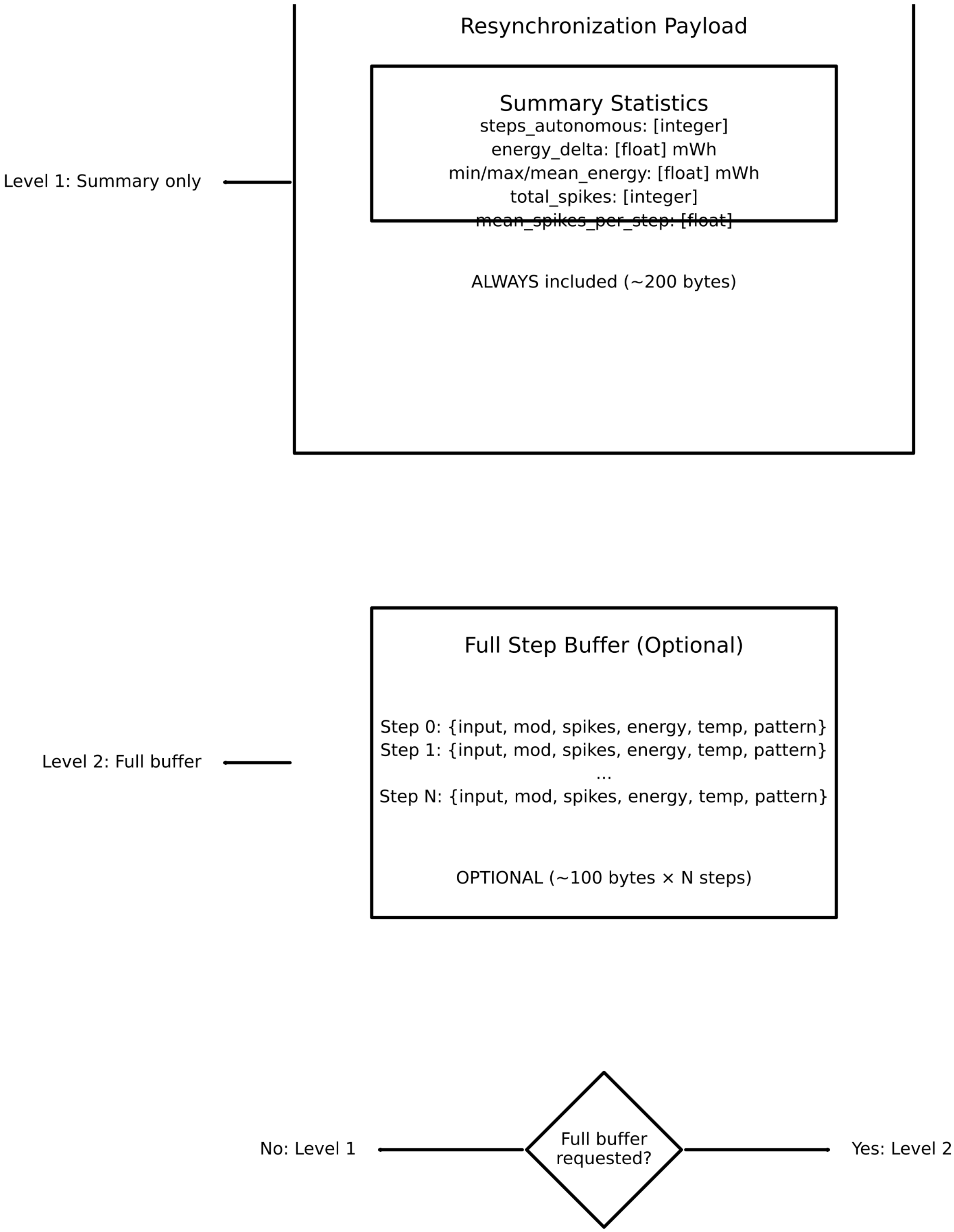

Resynchronization Payload
Summary Statistics
steps_autonomous: [integer]
energy_delta: [float] mWh
min/max/mean_energy: [float] mWh
total_spikes: [integer]
mean_spikes_per_step: [float]
ALWAYS included (~200 bytes)
Level 1: Summary only
Full Step Buffer (Optional)
Step 0: {input, mod, spikes, energy, temp, pattern}
Step 1: {input, mod, spikes, energy, temp, pattern}
...
Step N: {input, mod, spikes, energy, temp, pattern}
OPTIONAL (~100 bytes × N steps)
Level 2: Full buffer
Full buffer requested?
No: Level 1
Yes: Level 2

FIG. 5

Timeline A — Normal Reconnection:

-- Connected --	-- Timeout --	-- Autonomous --	-- Resync --	-- Connected -->
[Claude active]	[Watchdog counting]	[Self-regulated]	[Buffer sent]	[Control restored]

Timeline B — Crash Recovery:

-- Connected --	-- CRASH --	-- Restart --	-- Snapshot --	-- Connected -->
[Normal operation]	[Server terminates]	[Process relaunched]	[Reads latest.json]	[State restored]

Max state loss = 50 steps (snapshot interval)

Timeline C — Clean Shutdown:

-- Connected --	-- Autonomous --	-- EOF Signal --	-- Shutdown
[Normal operation]	[Fallback operation]	[stdin closed]	[Final snapshot written]

State preserved for next session — zero data loss

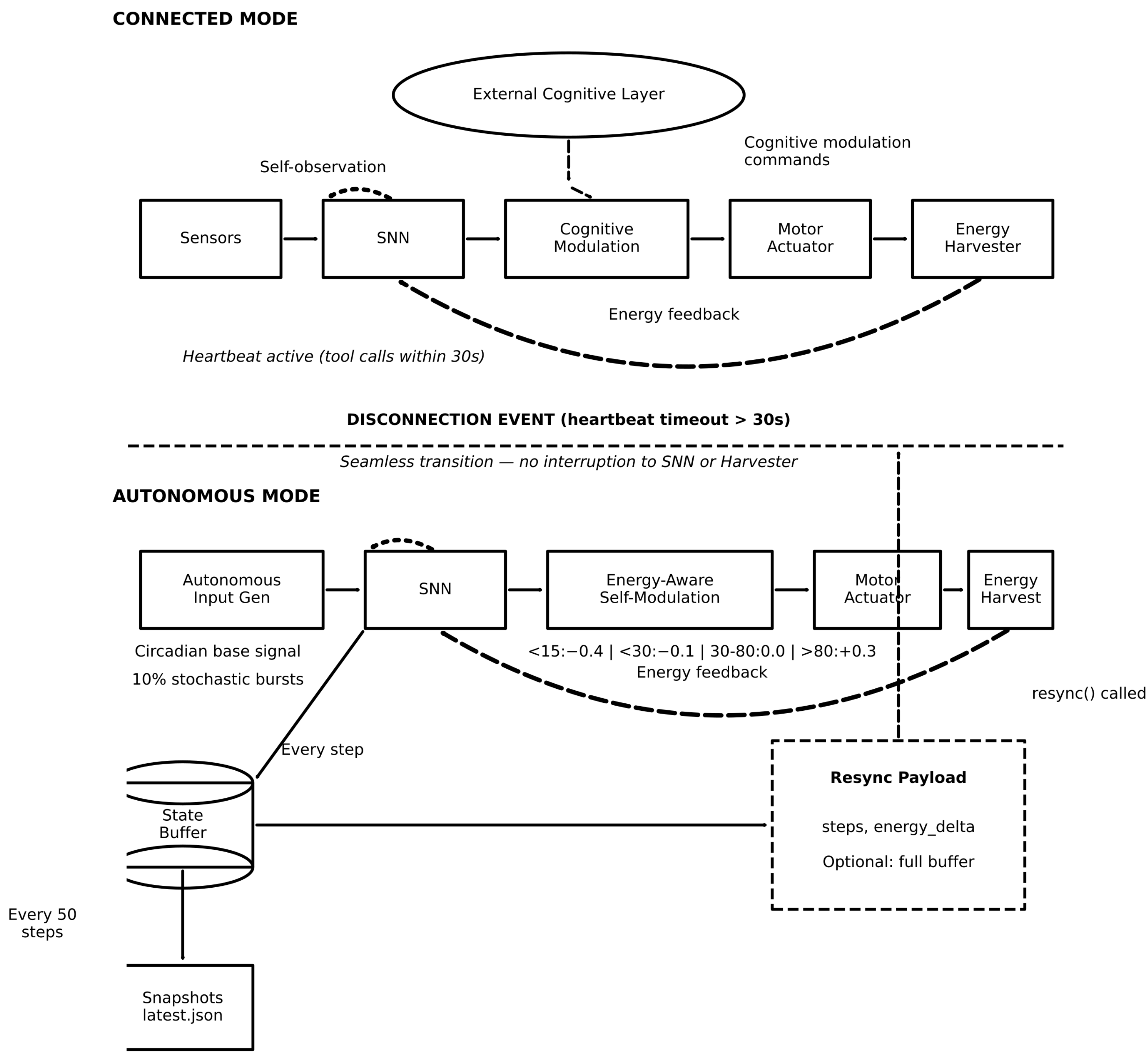
Solid segments: System actively processing

Dashed segments: System in transition

Bold marker at each state change: No data loss at any transition

FIG. 6

End-to-End Signal Flow: Connected vs. Autonomous Operation



The fallback system changes the INPUT SOURCE and MODULATION SOURCE

The core neural processing, motor actuation, and energy harvesting

continue WITHOUT INTERRUPTION through the transition

FIG. 7

Forward signal Energy feedback Self-observation